

$k[x]$ -modules & k -pairs

Notation: "Module over R "

||

" R -module"

Proposition

$$\left\{ \begin{array}{l} k\text{-pairs} \\ (V, T) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} k[x] \\ \text{-modules} \end{array} \right\}$$

Pf. If M is a $k[x]$ -module

then we have the k -pair

$$\left(\underset{\uparrow}{M}, \underset{\uparrow}{T} \right)$$

"remembers"

only scalar
mult. by k

multiplication
by x

i.e. $T(m) = x \cdot m \quad \forall m \in M$
(*)

Need to verify that T is
 k -linear

$$\begin{aligned} (a) \quad T(m_1 + m_2) &= T(m_1) + T(m_2) \\ || \quad \text{property (a)} \quad || \\ x \cdot (m_1 + m_2) &\stackrel{d}{=} x \cdot m_1 + x \cdot m_2 \end{aligned}$$

$$(b) \quad \forall a \in k, m \in M$$

$$\begin{aligned} T(a \cdot m) &= a \cdot T(m) \\ || \quad \quad \quad a \cdot (x \cdot m) \quad || \end{aligned}$$

$$x \cdot (a \cdot m) = (x \cdot a) \cdot m = (a \cdot x) \cdot m$$

$k[x]$ is a commutative ring
property (d)

R_{mk} , k : a field

$$\{k\text{-vector space}\} \leftrightarrow \{k\text{-module}\}$$

Other direction,

$$(V, T) : k\text{-pair}$$

$\downarrow$

We will produce a ring homomorphism

$$k[x] \longrightarrow \text{End}(V)$$

$$p(x) \longmapsto p(T)$$

HW 6, problem ①

$$1 \nmid p(x) = a_n x^n + \dots + a_1 x + a_0$$

$$p(T)(v) = a_n T^n(v) + \dots + a_1 T(v) + a_0 v$$

This gives $V \subset K[x]$ -module structure where

$$p(x) \cdot v = p(T)(v)$$

$$v \in V$$

In particular, $x \cdot v = T(v)$ (**)

$$\left\{ \begin{array}{l} K[x] \\ \text{-modules} \end{array} \right\} \longrightarrow \left\{ K\text{-pairs} \right\}$$

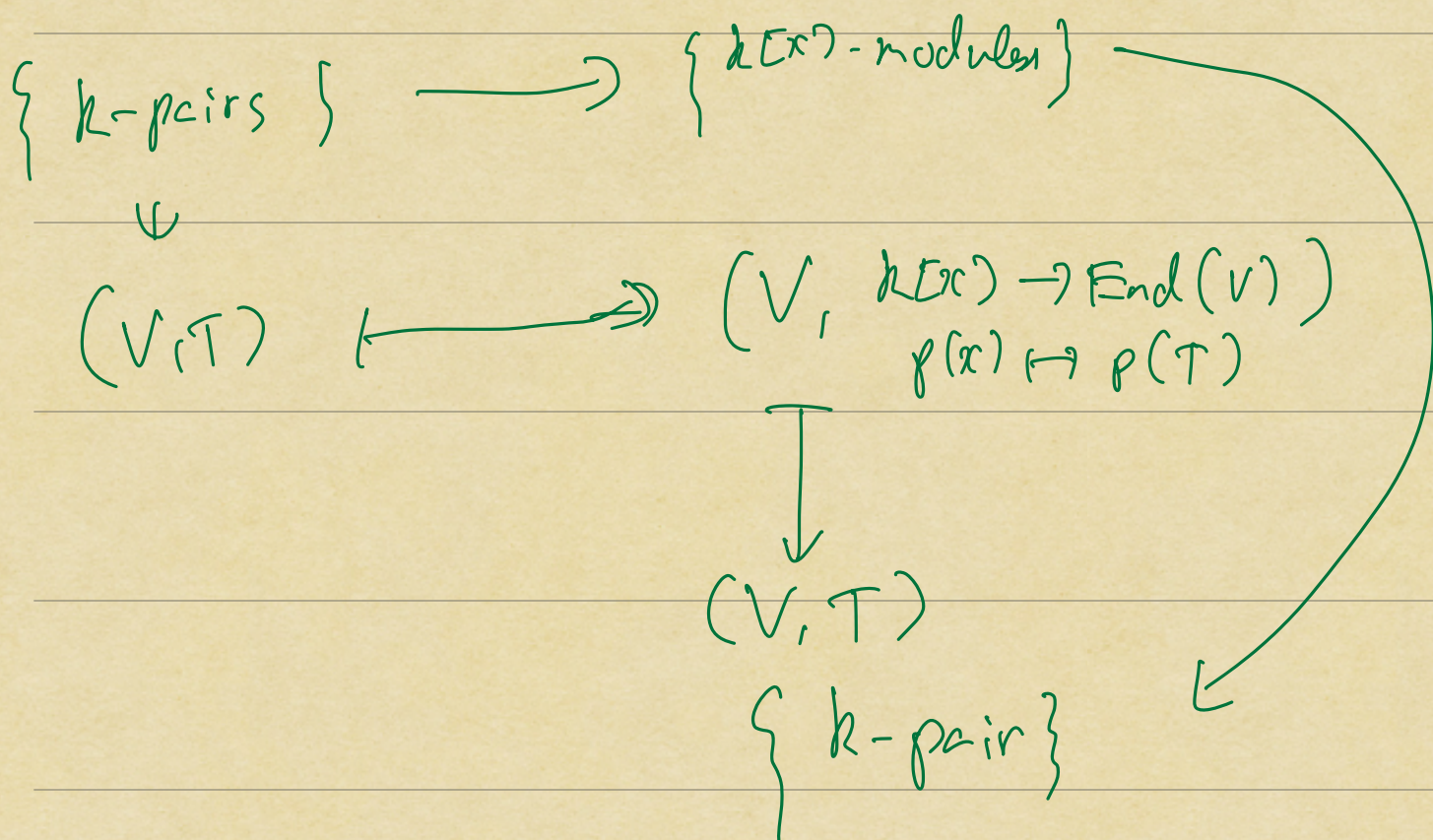
$$\begin{array}{c} \psi \\ M \end{array} \longmapsto (M, T)$$

↑
multiplication
by x

$\downarrow$
 M'

$\left\{ \begin{array}{l} k[x] \\ \text{-modules} \end{array} \right\}$

In fact $M = M'$ as $k[x]$ -modules



Defⁿ If M_1 & M_2 are
two R -modules, an

R -linear map or homomorphism

$$f: M_1 \longrightarrow M_2$$

is a group homomorphism s.t.

$\forall m \in M_1$ and $r \in R$

$$f(r \cdot m) = r \cdot f(m) \in M_2$$

Rmk. ① $R = \mathbb{Z}$

$$\{\mathbb{Z}\text{-modules}\} = \left\{ \begin{array}{l} M: \text{abelian gr} \\ + \\ \psi: \mathbb{Z} \rightarrow \text{End}(M) \\ \text{ring hom.} \end{array} \right\}$$

↑

this is unique!
 $\psi(a) = a \cdot \text{id}_M$

$$= \{ \text{Abelian groups} \} !$$

$$\{ \mathbb{Z} \text{-module maps} \} \longleftrightarrow \{ \text{Abelian group homomorphisms} \}$$

(2) k : field

$$\{ k\text{-linear transformations} \} \longleftrightarrow \{ k\text{-linear maps} \}$$

(3) $R = k[x]$

• An isomorphism of k -pairs

$$(V_1, T_1) \xrightarrow{\sim} (V_2, T_2)$$

is a k -linear isomorphism

$$\varphi: V_1 \longrightarrow V_2$$

s.t.

$$\begin{array}{ccc} V_1 & \xrightarrow{\varphi} & V_2 \\ T_1 \downarrow & & \downarrow T_2 \\ V_1 & \xrightarrow{\varphi} & V_2 \end{array}$$

$$T_2 \circ \varphi = \varphi \circ T_1$$

$$\Leftrightarrow$$

$$T_2 = \varphi \circ T_1 \circ \varphi^{-1}$$

$$\left\{ \begin{array}{l} k[x] - \text{linear} \\ \text{isomorphisms} \end{array} \right\} \quad (\longrightarrow) \quad \left\{ \begin{array}{l} \text{isomorphisms} \\ \text{of } k\text{-pairs} \end{array} \right\}$$